>_ CYBER TECH ACADEMY

Networking Fundamentals

Understand IP addressing, ports, DNS, routing, HTTP, troubleshooting, and packet analysis.

LEVEL: BEGINNER

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How to use this guide

Read the concepts first, run examples only in a system you own or are authorised to use, complete the lab, then answer the review questions without looking back. Commands and outputs can differ by Linux distribution, software version, and local configuration.

Responsible learning: only test systems for which you have explicit permission. These notes are educational and do not grant permission to access, scan, disrupt, or modify third-party systems.

Learning objectives

- Describe common network devices and network types.
- Explain IPv4, IPv6, ports, TCP, and UDP.
- Describe DNS and the path from a browser to a website.
- Use basic diagnostic commands on networks you administer.
- Interpret simple packet and subnet examples.

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1. Network building blocks

A computer network connects devices so they can exchange data. A LAN usually covers a limited local area; a WAN connects networks across larger distances; WLAN refers to a wireless local area network. A switch connects devices within a local network, a router forwards traffic between networks, an access point provides wireless connectivity, and a firewall applies traffic rules.

An IP address identifies a network interface at the network layer. A MAC address is a link-layer identifier used on a local network segment. They serve different purposes and are not interchangeable. IPv4 addresses are 32 bits, commonly written as four decimal octets; IPv6 addresses are 128 bits and use hexadecimal notation.

Private IPv4 ranges commonly used inside networks are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16. Private addresses are not directly routable across the public internet without translation or other gateway mechanisms.

2. TCP, UDP, ports, and services

TCP provides a reliable, ordered byte stream with connection management and retransmission. UDP sends datagrams without TCP's built-in reliability guarantees; it can be useful where low overhead or application-controlled timing matters. A port identifies an application endpoint on a host. An open port indicates a service is accepting or listening for traffic, not automatically that it is vulnerable.

Port	Typical use	Learning point
22/TCP	SSH	Secure remote administration; protect keys and access.
53	DNS	Name resolution; may use UDP or TCP.
80/TCP	HTTP	Web traffic without TLS encryption.
123/UDP	NTP	Time synchronisation.
443/TCP	HTTPS	HTTP protected with TLS.
445/TCP	SMB	File sharing; restrict exposure.
3389/TCP	RDP	Remote desktop; restrict and protect access.

Services can use different ports, and administrators can configure non-default ports. A port number alone is not proof of the service or its security state. Confirm findings with authorised evidence.

3. DNS and web requests

DNS translates domain names into records such as IPv4 A records and IPv6 AAAA records. Other records include MX for mail routing, CNAME for aliases, and TXT for text-based data used for verification and email-security policies. Resolvers may cache answers according to TTL values.

```
getent hosts example.com
# If installed:
dig example.com A
dig example.com MX
```

When a browser opens a website, it typically resolves the name, establishes network connectivity, negotiates TLS for HTTPS, sends an HTTP request, and receives a response. HTTP methods include GET and POST; common status

codes include 200 (success), 301/302 (redirect), 400 (bad request), 403 (forbidden), 404 (not found), and 500 (server error).

```
curl -I https://example.com
```

HTTPS protects data in transit using TLS, provided certificate validation succeeds and the connection is configured correctly. HTTPS does not guarantee that a site is honest or that its application is free of vulnerabilities.

4. Subnetting and routing

Subnetting divides an IP address space into smaller networks. CIDR notation places the prefix length after a slash: /24 means 24 bits are the network prefix in IPv4, leaving 8 host bits. A traditional /24 has 256 total addresses; in a typical IPv4 LAN, the network and broadcast addresses are not assigned to hosts, leaving 254 conventional usable host addresses.

CIDR	Total IPv4 addresses	Typical usable hosts
/24	256	254
/25	128	126
/26	64	62
/27	32	30
/28	16	14

A routing table tells a host where to send packets for different destinations. The default route is used when no more specific route matches. A gateway commonly forwards traffic beyond the local subnet.

```
ip addr
ip route
ip -6 route
```

IPv6 has no broadcast addressing; it uses multicast and other mechanisms. Do not apply IPv4 broadcast-address assumptions to IPv6.

5. Troubleshooting and packet analysis

Troubleshooting should follow evidence rather than guesses. Check the local interface and IP configuration, verify the expected gateway and routes, test reachability where permitted, inspect DNS, and finally inspect the application response. A failed ping is not conclusive proof that a host is down because ICMP may be filtered.

```
ip addr
ip route
ping -c 4 192.168.1.1
getent hosts example.com
curl -I https://example.com
```

Packet analysis examines captured network frames. Wireshark can display protocols, addresses, ports, timestamps, and request/response sequences. Use sample captures or capture only traffic you own or have explicit permission to inspect. Captures can contain credentials, personal data, and confidential content; handle them securely.

Useful concepts include the TCP three-way handshake (SYN, SYN-ACK, ACK), DNS query/response pairs, and the distinction between packet headers and application payload. With HTTPS, application content is encrypted in transit, though some metadata may remain visible.

Guided practical lab

Goal: diagnose a simulated connectivity issue using a repeatable sequence. Run diagnostics only on your own device or authorised lab.

- Record the interface and IP address using ip addr.
- Inspect routes using ip route and identify the default gateway if present.
- Test the gateway only if you know its address and are allowed to do so.
- Resolve a domain using getent hosts example.com.
- Inspect the HTTPS response using curl -I https://example.com.
- Write down which layer each result helps you investigate.

Do not scan public IP ranges or other people's devices without explicit authorisation. This lab is about local diagnosis, not unauthorised discovery.

Knowledge check

Try to answer these before checking the answer key. Write your reasoning, not just a single word.

1. How does a router differ from a switch?

Answer: _____

2. Does an open port prove a vulnerability?

Answer: _____

3. What is DNS used for?

Answer: _____

4. What does /26 mean in IPv4 CIDR notation?

Answer: _____

5. Why might ping fail even when a server is available?

Answer: _____

Answer key and reasoning

1. A switch commonly forwards frames within a local network; a router forwards packets between IP networks using routing information.

2. No. It indicates a service is listening or accepting traffic; vulnerability depends on service configuration, software, exposure, and other controls.

3. DNS maps names to records such as IP addresses and mail-routing information.

4. The first 26 bits are the network prefix, leaving 6 host bits and 64 total addresses in that block.

5. ICMP may be filtered or deprioritised; test the actual service and other evidence too.

Glossary

Term	Meaning
IP address	Network-layer address for an interface.
Port	Application endpoint identifier on a host.
DNS	System for resolving domain names and other records.
CIDR	Notation specifying a network prefix length.
Gateway	Device or route used to reach another network.
Packet capture	Recorded network traffic used for analysis.

Further study

Continue with the next Cyber Tech Academy module, practise in a disposable virtual machine or an explicitly authorised lab, and consult official manuals for the tools you use. Verify commands against your own system before running them.

Cyber Tech Academy • Free learning • Learn the theory. Practise safely. Build real skills.